HCP 1033 (cpc_1906-08-23_DE_Munsterlager_DE_Ahlden_): a soldier's number-cipher card to Hertha, 1906

Human solver: Tomáš Pavuk

Status: deciphered in full (330 of 330 cipher numbers). The independent review 1 verdict was ACCEPT, and the decoded text did not change. Report dated 2026-11-03.

Origin and context

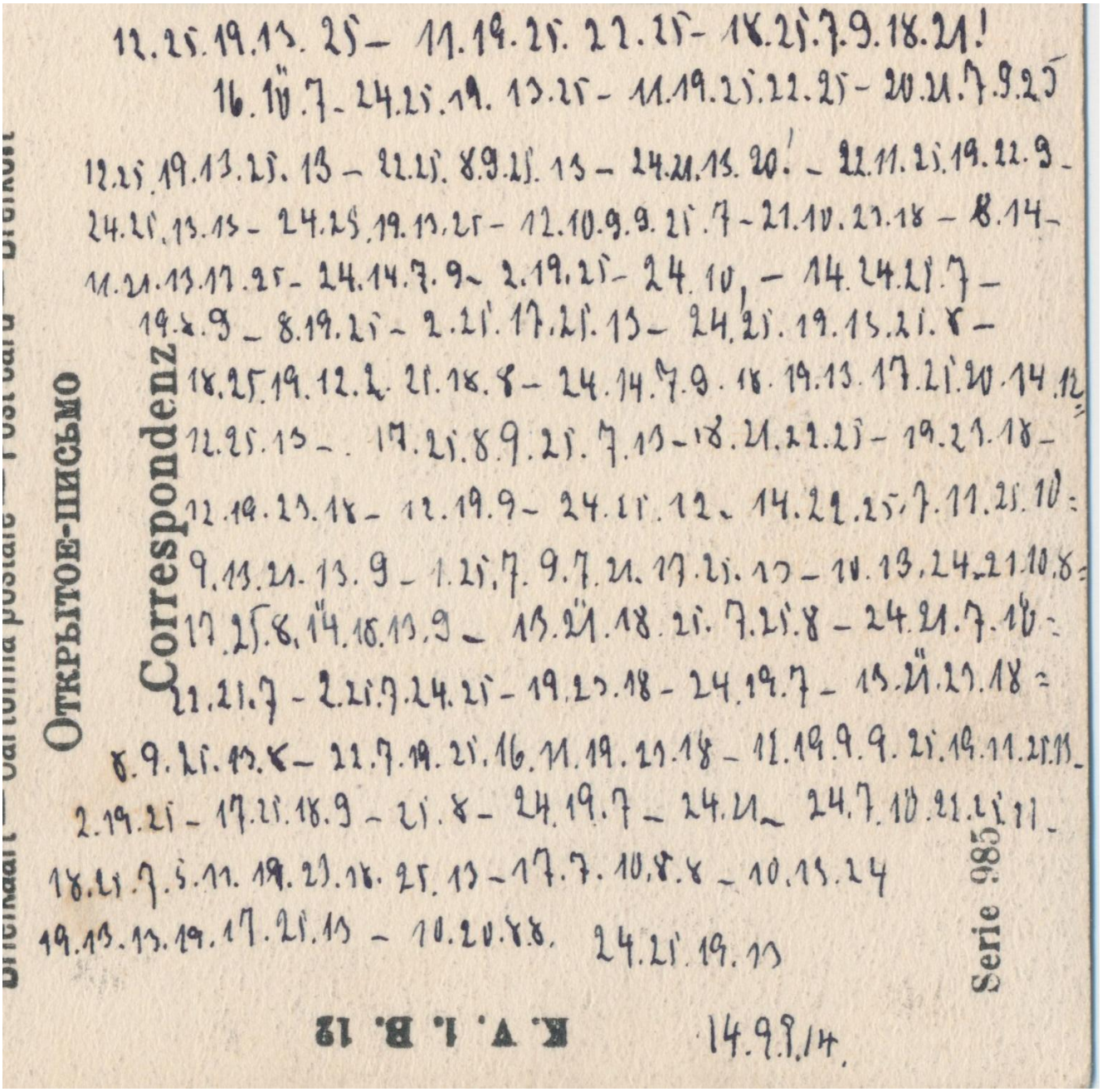
The cryptogram is record 1033 of the HC Portal (https://crypto.hcportal.eu/dashboard/cryptograms/1033), portal name cpc_1906-08-23_DE_Munsterlager_DE_Ahlden_. It belongs to Tobias Schrödel's collection of cryptographic postcards (private collection). When the record was captured on 2026-11-03, the portal listed it as **Not solved**. It was tagged Postcard and T. Schrödel Collection, with language Unknown, sender Unknown and recipient "Hertha Tesohe". No transcription or solution was attached. The portal's only reference is Schrödel's survey paper "Cryptographic postcards" (HistoCrypt 2021).

The picture side shows a girl with flowers in her hair and carries no writing. The address side has a German 5-pfennig Germania stamp. Its cancellation reads **MUNSTERLAGER (BZ. HANNOVER), 23.8.06**. The plain Kurrent address reads Fräulein Hertha Tesche (the portal has "Tesohe"), per Adr. Herrn Gastwirth(?) Koch, Ahlden a/Aller. The surname and the occupation word are uncertain, and they are not part of the cipher. Munsterlager was the large army training ground on the Lüneburg Heath, about 50 km from Ahlden.

The whole message is in numbers, written along the long edge of the address side, so the card must be turned 90 degrees to read it. There are 16 lines of text, then the signature "14.9.9.14". A two-line postscript is written upside down above the printed heading.



Address side as scanned (source/address-large.jpg)



Message area turned upright (work/crops/body_rot90.jpg)

Method

The transcription was made by hand, number by number, from enlarged crops of the full-resolution scan. Numbers are separated by dots. Words are separated by a dash. A "=" or ":" at the end of a line marks a word broken across lines. Two dots above a number mark an umlaut. No transcription was supplied, and no OCR was used.

The key came from the opening line, "12.25.19.13.25 - 11.19.25.22.25 - 18.25.7.9.18.21 !". This has the shape of a salutation, and with the recipient's name as a crib it reads MEINE LIEBE HERTHA! The values recovered from it showed a simple rule. The rest of the key followed from that rule, and every other word on the card confirmed it.

Key

The cipher is a monoalphabetic number substitution (register type MASC). The 25-letter alphabet without J is cut into five blocks of five letters. The blocks are numbered in reverse order, and the letters inside each block keep their normal order.

numbers	1	2	3	4	5
21-25	A	B	C	D	E
16-20	F	G	H	I	K
11-15	L	M	N	O	P
6-10	Q	R	S	T	U
1-5	V	W	X	Y	Z

On the card, 21 distinct numbers occur. Four values are not seen on the card: 3 (X), 4 (Y), 6 (Q) and 15 (P). They are filled in from the block rule only. Ä, Ö and Ü are written as 21, 14 and 10 with two dots above.

Transcription

```
12.25.19.13.25 - 11.19.25.22.25 - 18.25.7.9.18.21 !
16.10".7 - 24.25.19.13.25 - 11.19.25.22.25 - 20.21.7.9.25
12.25.19.13.25.13 - 22.25.8.9.25.13 - 24.21.13.20 ! - 22.11.25.19.22.9 -
24.25.13.13 - 24.25.19.13.25 - 12.10.9.9.25.7 - 21.10.23.18 - 8.14 -
11.21.13.17.25 - 24.14.7.9 - 2.19.25 - 24.10 , - 14.24.25.7 -
19.8.9 - 8.19.25 - 2.25.17.25.13 - 24.25.19.13.25.8 -
18.25.19.12.2.25.18.8 - 24.14.7.9.18.19.13.17.25.20.14.12 =
12.25.13 - . 17.25.8.9.25.7.13 - 18.21.22.25 - 19.23.18 -
12.19.23.18 - 12.19.9 - 24.25.12 - 14.22.25.7.11.25.10 =
9.13.21.13.9 - 1.25.7.9.7.21.17.25.13 - 10.13.24 - 21.10.8 =
17.25.8.14".18.13.9 - 13.21".18.25.7.25.8 - 24.21.7.10" =
22.{25}.7 - 2.25.7.24.25 - 19.23.18 - 24.19.7 - 13.21".23.18 =
8.9.25.13.8 - 22.7.19.25.16.11.19.23.18 - 12.19.9.9.25.19.11.25.13 -
2.19.25 - 17.25.18.9 - 25.8 - 24.19.7 - 24.21 - 24.7.10".{22}.25.{13} -
18.25.7.5.11.19.23.18.25.13 - 17.7.10.8.8 - 10.13.24
19.13.13.19.17.25.13 - 10.20.8.8 - 24.25.19.13
14.9.9.14 .
postscript, upside down:
17.7.10.8.8.25 - 22.19.9.9.25 - 24.25.19.13.25 - {12}.10.9 :
9.{25}.7 - 1.14.13 - 12.19.7 -
```

A trailing " marks two umlaut dots. A number in braces {n} has an ambiguous shape and is read as n by word sense (see Uncertainties).

Plaintext

Decoded, line by line:

```
meine liebe hertha!
für deine liebe karte
meinen besten dank! bleibt
denn deine mutter auch so
lange dort wie du, oder
ist sie wegen deines
heimwehs dorthingekom=
men gestern habe ich
mich mit dem oberleu=
tnant vertragen und aus=
gesöhnt näheres darü=
ber werde ich dir näch=
stens brieflich mitteilen
wie geht es dir da drüben
herzlichen gruss und
innigen ukss dein
otto
[postscript]
grusse bitte deine mut=
ter von mir
```

Normalised German:

Meine liebe Hertha! Für deine liebe Karte meinen besten Dank! Bleibt denn deine Mutter auch so lange dort wie du, oder ist sie wegen deines Heimwehs dorthin gekommen? Gestern habe ich mich mit dem Oberleutnant vertragen und ausgesöhnt; Näheres darüber werde ich dir nächstens brieflich mitteilen. Wie geht es dir da drüben? Herzlichen Gruß und innigen Kuss, dein Otto. P.S. Grüße bitte deine Mutter von mir.

English:

My dear Hertha! Many thanks for your dear card! Is your mother staying there as long as you, or did she come there because of your homesickness? Yesterday I made up and was reconciled with the first lieutenant; I will tell you more about it soon by letter. How are you over there? Warm greetings and a fond kiss, your Otto. P.S. Please give my regards to your mother.

Slovak:

Moja milá Hertha! Za tvoju milú pohľadnicu ti zo srdca ďakujem! Zostane tvoja mama tam tak dlho ako ty, alebo tam prišla kvôli tvojej túžbe po domove? Včera som sa s nadporučíkom udobril a zmieril; podrobnosti ti čoskoro napíšem v liste. Ako sa ti tam darí? Srdečný pozdrav a vrúcny bozk, tvoj Otto. P.S. Pozdravuj, prosím, odo mňa mamu.

Reproduction

From the repository root:

```
python3 -I ciphers/hcp-1033/work/decode.py
```

The script checks the SHA-256 of the four scan files, prints the key and decodes the transcription line by line.

Uncertainties

None of these changes the sense of the letter. Each is a single sign inside a word that otherwise reads.

1. "10.20.8.8" decodes to UKSS. Both numbers are clearly 10 and 20, so the writer swapped the first two numbers of KUSS. "Kuss" is an editorial correction.
2. The writer usually keeps 25 (a "2" with a hooked, i-like "5") apart from 21 (a "2" with a plain "1"). Two places are read by word sense: (darü)BER on line 12 and (Mut)TER in the postscript, where the middle number looks closer to 21 than to 25.
3. In the postscript, MUT has a first number that looks like 11 (L). It is read as 12 (M) by sense.
4. DRÜBEN on line 14. The 4th number is read as 22 (B). The reviewer finds it fairly clear, while the solver sees its second digit as closer to a "1". The last number is cramped at the card edge and looks like 11. It is read as 13 (N) by sense ("wie geht es dir da drüben").
5. The postscript's first word may be GRUSSE or GRÜSSE. A possible umlaut dot merges with the printed letters of the heading.
6. The signature "14.9.9.14" looks like a date, but it decodes cleanly to OTTO and stands where a name is expected after "dein".
7. Under the stamp. The stamp is intact on the scan, so what lies under it cannot be checked. The picture side and the margins show no further writing, and nothing visible points to a hidden message.
8. The address (Tesche or Tesohe, Gastwirth?) is plain writing, not cipher, and is read only tentatively.

Review summary

An independent reviewer (a separate Claude Opus 5.5 agent) did the following:

- ran the reproduction command (all hash checks OK)

- decoded the transcription with its own script and a literally written key, and got the same text
- re-read every line against its own enlargements of the scan
- checked the picture side and the stamp box for hidden writing

Its verdict was **ACCEPT**, with four minor findings:

- The word-sense calls were all already disclosed. The reviewer read the 22 in DRÜBEN as clearer than the solver had. The solver kept its cautious mark, and both views are given above.
- Check whether the cited HistoCrypt paper prints this card. The solver downloaded the full paper and searched it. It is a survey and contains nothing about this card (no Munsterlager, Ahlden, Hertha or Otto; its only 1906 card was sent from Finland to Russia).
- Present the four unused key values as rule-derived. This is done in the Key section.
- GRUSSE versus GRÜSSE remains open.

The decoded text did not change because of the review.

Because similar cards from another series carry steganographic messages, the card was also checked for hidden writing. Nothing was found. The type is therefore MASC, not "MASC + Steganography".

Sources

- HC Portal record 1033 and its scans: https://crypto.hcportal.eu/dashboard/cryptograms/1033 (picture and address side; full-resolution originals from api.hcportal.eu media 3159 and 3160).
- T. Schrödel, "Cryptographic postcards", HistoCrypt 2021, https://ecp.ep.liu.se/index.php/histocrypt/article/view/166 (checked; it holds no solution of this card).
- Transcription, key recovery and decipherment: Claude Opus 5.5, 2026-11-03. Independent review: a separate Claude Opus 5.5 agent.